Vinayak Kamat

Machine Learning Engineer | Production ML, Model Monitoring and Financial AI

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Summary

AI/ML Engineer with 1 year 8 months at KAN Innovations, now owning the deployment and monitoring of production ML models. Experience with PyTorch, model calibration, ONNX inference optimization, MLflow-based tracking, drift detection, and AWS deployment. Built Aethelgard, a real-time gateway that scores LLM outputs for risk and streams metrics to Datadog. Pursuing an MSc in Financial Engineering.

Experience

KAN Innovations, AI/ML Engineer — Mumbai, Maharashtra
July 2026 – present
4 months

- Own the ML project end to end, including deployment and ongoing monitoring of ML models and agents in production
- Improved model latency for production inference, including converting PyTorch models to ONNX
- Promoted from Junior AI/ML Engineer after 14 months based on performance

KAN Innovations, Junior AI/ML Engineer — Mumbai, Maharashtra
May 2025 – June 2026
1 year 2 months

- Integrated ML models into production pipelines and deployed them on AWS (Lambda, ECR, S3, DynamoDB)
- Improved model calibration so prediction confidence better reflects real accuracy, making production predictions more reliable
- Evaluated models and monitored them for drift, and tracked experiments and model versions with MLflow
- Built and evaluated Retrieval-Augmented Generation (RAG) pipelines, measuring output quality across retrieval and generation

KAN Innovations, ML Engineer Intern — Mumbai, Maharashtra
Feb 2025 – May 2025
3 months

- Built a PyTorch ulcer detection model reaching about 85% accuracy
- Improved the performance of an eye posture detection model using OpenCV

Projects

Aethelgard

Real-time AI governance gateway that scores LLM outputs for hallucination and risk.

- Built a FastAPI gateway that analyzes model outputs in real time, using a secondary model for semantic risk scoring
- Streamed risk metrics to Datadog for continuous monitoring of model output quality
- Designed automated detection logic to flag unreliable outputs before they reach users

Financial AI Agent (Demo)

Prototype agent for financial data retrieval and reasoning in Python.

- Built modular retrieval and reasoning pipelines for financial insights
- Implemented query routing and response refinement in Python

Skills

ML and Frameworks: Python, PyTorch, Transformers, spaCy, OpenCV, ONNX, model calibration

MLOps and Monitoring: MLflow (experiment tracking, model versioning), model monitoring and drift detection, Datadog, Docker

Backend and Cloud: FastAPI, Flask, REST APIs, AWS (Lambda, ECR, S3, DynamoDB)

Data: Pandas, NumPy

Education

MSc **WorldQuant University**, Financial Engineering — Remote, Washington DC
Sept 2025 – May 2027

B.Tech **Punjab Technical University**, Computer Science and Engineering — Punjab, India
Sept 2021 – May 2025

Achievements

Qualified the Graduate Aptitude Test in Engineering (GATE)